Module 4 — Quiz A (Def & Notation)

Question bank · After Read 1 · open notes · unlimited attempts

Module 4, Quiz A: Definitions & notation

Reading: Module 4 Read 1
Project: Project 4 · **Canvas:** Def & Notation quiz (Module 4)

Work these **practice** items after Read 1. Sections follow Read 1: study-design vocabulary (EV, RV, RCT, observational, confounder, DAG) and causal language.

Methods-map matching and R output → Quiz B after Read 2.

Section A — Study-design vocabulary (Read 1 §1)

A1. Explanatory variable

Define **explanatory variable** (EV) in one or two clear sentences.

A2. Response variable

Define **response variable** (RV) in one or two clear sentences.

A3. Observational study

Define **observational study** and distinguish it from a randomized experiment.

A4. Randomized controlled trial

Define **randomized controlled trial (RCT)** and state its key feature.

A5. Confounder

Define **confounder** in one sentence. Give one example.

A6. DAG

Draw or describe a **DAG** (directed acyclic graph) with one confounder C affecting both X (EV) and Y (RV).

A7. p-value limitation in observational studies

State what a **p-value does NOT** establish for a two-group difference in an observational study.

A8. Random assignment benefit

Explain in one sentence why **random assignment** is the key tool for causal inference.

Answer key — Section A — Study-design vocabulary (Read 1 §1)

1. The **explanatory variable** is the variable that explains, predicts, or is thought to cause changes in the response — typically the treatment, exposure, or grouping factor.
2. The **response variable** is the outcome of interest — the variable we measure to see whether it changes with the explanatory variable.
3. An **observational study** records data on individuals as they are, without the researcher assigning treatments. Group membership is determined by the individuals themselves, not by random assignment.
4. An **RCT** is an experiment in which participants are **randomly assigned** to treatment or control groups by the researcher, so group membership is not self-selected.
5. A **confounder** is a variable associated with both the EV and RV that can create or distort the apparent association between them. Example: age is a confounder in a study of coffee and heart disease.
6. $C \to X$ and $C \to Y$. This 'backdoor path' from X to Y through C creates confounded association if C is not controlled.
7. A p-value does **not** establish **causation**. Even a very small p-value only indicates the difference is unlikely under H_0; confounders can still explain an observational association.
8. Random assignment distributes known **and unknown** confounders approximately equally across groups on average, so observed group differences are attributable to the treatment.

Section B — Causal language vs association language (Read 1 §1–2)

B1. Causal vs association language

Classify each claim: **causal** (requires RCT) or **association only** (observational):

a) "Students who took the tutoring program **scored higher** on the exam."
b) "Students **randomly assigned** to tutoring **scored higher** on the exam."

B2. Subclassification

A measured confounder exists in an observational study. What technique can partially account for it, and what is its limitation?

B3. Identify the confounder

Study: older employees have higher salaries and are more likely to be managers. Is "age" a confounder for the association between managerial status and salary? Explain.

Answer key — Section B — Causal language vs association language (Read 1 §1–2)

1. a) **Association only** — students self-selected into tutoring; confounders likely. b) **Causal** — random assignment supports the tutoring → higher score claim.
2. **Subclassification / stratification**: compare groups within levels of the confounder. Limitation: only controls for **measured** confounders; unmeasured confounders remain.
3. **Yes.** Age is associated with both being a manager (EV) and salary (RV), so it can create or distort the managerial status → salary association.
